



BITS Pilani

Introduction to Statistical Methods

ISM Team



< ISM >

Lecture No. 6



Session 6

(22nd Feb, 2026)



Work Integrated Learning Programmes

Probability Distributions

Agenda



Distributions:-

- ❖ Bernoulli
- ❖ Binomial
- ❖ Poisson
- ❖ Normal distributions.
- ❖ Introduction to t-distribution
- ❖ Introduction to F-Distribution and
- ❖ Introduction to Chi Square distributions

Introduction



The Bernoulli distribution is a univariate discrete distribution used to model random experiments that have binary outcomes.

- ◆ Single random experiment
- ◆ Dichotomous outcome



Bernoulli Distribution



Definition

A random variable 'X' is said to have Bernoulli distribution if its probability mass function is given by

$$p(x) = \begin{cases} p^x q^{1-x}, & x = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

This represents a single trial with two possible outcomes (usually labeled as success/failure or 1/0) where success has a probability p and failure has a probability $q = 1 - p$.

Mean & Variance



The expected value of a Bernoulli random variable X is $E[X] = p$

Proof:

$$\begin{aligned} E[X] &= \sum_{x=0,1} xp(x) \\ &= 0 \cdot p(0) + 1 \cdot p(1) \\ &= 0 + 1 \cdot (p) \\ &= p \end{aligned}$$

The variance of a Bernoulli random variable X is $\text{Var}[X] = pq$

Proof:

$$\begin{aligned} E[X^2] &= \sum_{x=0,1} x^2 p(x) \\ &= 0^2 \cdot p(0) + 1^2 \cdot p(1) \\ &= 0 + 1 \cdot p \\ &= p \end{aligned}$$

$$\text{Var}[X] = E[X^2] - E[X]^2 = p - p^2 = p(1 - p) = pq$$



Binomial Distribution

- Binomial distribution is a discrete probability distribution.
- Binomial distribution will be applied under the following experimental conditions
 - 1) The number of trials (n) is finite
 - 2) The trials are independent of each other
 - 3) The probability of success p is constant for each trial.
 - 4) Each trial results in two mutually exclusive events known as success and failure.

Binomial Distribution



Definition

A random variable 'X' is said to have Binomial distribution if its probability mass function is given by

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & x = 0, 1, 2, 3, \dots, n \\ 0, & \text{elsewhere} \end{cases}$$

It describes the number of successes in a fixed number of independent Bernoulli trials. It involves parameters n (number of trials) and p (probability of success).

Binomial Distribution

n = # of trials

1-p = probability of failure

$${}^n C_x p^x (1-p)^{n-x}$$

x = # of success

p = probability of success

Mean & Variance



The expected value of a Binomial random variable X is $E[X] = np$

Proof:

$$\begin{aligned} E(x) &= E\left[\sum_{i=1}^n Y_i\right] \text{ (representation as a sum of } n \text{ independent Bernoulli r.v.)} \\ &= \sum_{i=1}^n E[Y_i] \text{ (linearity of the expected value)} \\ &= \sum_{i=1}^n p \text{ (expected value of a Bernoulli r.v.)} \\ &= np \end{aligned}$$

The variance of a Binomial random variable X is $\text{Var}[X] = npq$

Proof:

$$\begin{aligned} \text{Var}(x) &= \text{Var}\left[\sum_{i=1}^n Y_i\right] \text{ (representation as a sum of } n \text{ independent Bernoulli r.v.)} \\ &= \sum_{i=1}^n \text{Var}[Y_i] \text{ (linearity of Variance of independent variable)} \\ &= \sum_{i=1}^n pq \text{ (variance of a Bernoulli r.v.)} \\ &= npq \end{aligned}$$

Binomial Distribution



Which conditions for the binomial distribution, if any, fail to hold in the following situations?

(a) The number of persons getting Corona in a room of 30 persons.

Ans: Getting corona is not independent. So it is not binomial.

(b) Consider 2 out of 20 PCs are defective. We randomly select 3 for testing. Is this a binomial experiment?

Ans: If two defective pieces are in the first two selection, then the result of third is known already. Thus, the event is not independent. So this is not binomial.

Example 1:



In a particular restaurant, an average of 3 out of every 5 customers ask for water with their meal. A random sample of 10 customers is selected. Find the probability that

- (i) Exactly 6 ask for water with their meal,
- (ii) Less than 9 ask for water with their meal.
- (iii) No one asks for water with their meal?
- (iv) At most 2 ask for water with their meal?
- (v) At least 3 ask for water with their meal?

Solution:



$$n = 10, p = 3/5 = 0.6, q = 1 - p = 0.4$$

$$(i) p(6) = 0.251$$

$$(ii) p(\text{less than } 9) = 1 - [p(9) + p(10)] = 0.954$$

$$(iii) p(0) = 0.0001$$

$$(iv) p(\text{at most } 2) = p(0) + p(1) + p(2) = 0.0001 + 0.0016 + 0.0106 \\ = 0.0123$$

$$(v) p(\text{at least } 3) = p(3) + \dots + p(10) \\ = 1 - [p(0) + p(1) + p(2)] = 0.9877$$

Understandings



Phrase	<i>Math Symbol</i>
“at least”	$\geq$
“more than” or “greater than”	$>$
“fewer than” or “less than”	$<$
“no more than”	$\leq$
“exactly”	$=$

Example 2:



Ten coins are thrown simultaneously. what is the probability of getting 3 or fewer heads?

$$P(x \leq 3) = \sum p(x) = P(0) + P(1) + P(2) + P(3)$$

$$= {}^{10}C_0(0.5)^0(0.5)^{10} + {}^{10}C_1(0.5)^1(0.5)^9 + {}^{10}C_2(0.5)^2(0.5)^8 + {}^{10}C_3(0.5)^3(0.5)^7$$

$$= (0.5)^{10} [{}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3]$$

$$= (0.5)^{10} [1 + 10 + 45 + 120] = 176/1024 = 0.1718$$

$$P(x \leq 3) = \sum p(x) = 0.1718$$

Example 3:



A gun is fixed at a target from a certain distance. The probability of hitting the target is 0.2. Totally two bombs are enough to destroy the target. If six shells are aimed at the target, find the probability of the target being destroyed.

Solution: $p = 0.2$, $q = 0.8$, $n = 6$

$X \rightarrow$ number of shells required to destroy the target.

$$P(X \geq 2) = 1 - P(X < 2)$$

$$= 1 - [P(X=0) + P(X=1)]$$

$$= 0.3447$$

Example 4:



Each of five questions on a multiple-choice examination has four choices, only one of which is correct. The student is attempting to guess the answers. The random variable X is the number of questions answer correctly. What is the probability that the student will get

- a) exactly three correct answers?
- b) at most three correct answers?
- c) at least one correct answer?
- d) Find Mean and variance.

Example 5:



- Find the binomial distribution if the mean is 6 and variance is 4



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Example 6:



In a large number of parts manufactured by a machine, the mean number of defectives in a sample of 20 is 2. Out of 1000 such samples, how many would be expected to contain at least 3 defective parts.

Sol: Given $n=20$;

$$\text{mean} = \mu = 2, \text{ but } \mu = np \Rightarrow 2 = 20p \therefore p = 0.1$$

$$\therefore P(x) = {}^{20}C_x (0.1)^x (0.9)^{20-x}$$

- Therefore the probability that there are at least 3 defective parts in a sample of 20 is

$$P(X \geq 3) = 1 - P(X < 3) = 1 - [P(0) + P(1) + P(2)]$$

$$\therefore P(X \geq 3) = 0.323$$

- Hence out of 1000 samples we get $0.323 \times 1000 = 323$ samples

Binomial Distribution Recall



A **binomial experiment** is a probability experiment that satisfies the following conditions.

1. The experiment is **repeated** for a fixed number of trials, where each trial is **independent** of other trials.
2. There are only **two possible outcomes** of interest for each trial. The outcomes can be classified as a success (S) or as a failure (F).
3. The probability of a success $P(S)$ is the same for each trial.
4. The random variable x counts the number of successful trials.

Poisson Distribution



- Describes rare events
- Poisson distribution is suitable for rare events for which the probability of occurrence 'p' is very small and the number of trials 'n' is very large.
- Also binomial distribution can be approximated by Poisson distribution when
$$n \rightarrow \infty \text{ and } p \rightarrow 0 \text{ such that } \lambda = np = \text{constant}$$

Poisson Distribution



Definition

A random variable 'X' is said to have Poisson distribution if its probability mass function is given by

$$p(x) = \begin{cases} e^{-\lambda} \frac{\lambda^x}{x!}, & x = 0, 1, 2, \dots \\ 0, & \text{elsewhere} \end{cases}$$

- Poisson distribution is the discrete probability distribution of a discrete random variable X, which may not have upper bound. It is defined for non negative values of x .

Example 6:



Telephone calls enter a college switchboard on the average of two every three minutes. What is the probability of 5 or more calls arriving in a 9-minute period?

$$\lambda = 2 \cdot 9/3 = 6, \quad P(X \geq 5) = 1 - P(X \leq 4) \doteq 0.715$$

Example 7:



If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective.

Solution:

The probability of defective units $p = 3/100 = 0.03$

Give $n = 200$.

We observe that p is small and n is large here. Thus it is a Poisson distribution.

Mean $\lambda = np = 200 \times 0.03 = 6$

$P(X = x)$ is given by the Poisson Distribution Formula as $(e^{-\lambda} \lambda^x)/x!$

$P(X < 2) = P(X = 0) + P(X = 1)$

$= (e^{-6} 6^0)/0! + (e^{-6} 6^1)/1!$

$= e^{-6} + e^{-6} \times 6$

$= 0.00247 + 0.0148$

$P(X < 2) = 0.01727$

Example 8:



A computer crashes on average once every 4 months;

- a) What is the probability that it will not crash in a period of 4 months?
- b) What is the probability that it will crash once in a period of 4 months?
- c) What is the probability that it will crash twice in a period of 4 months?
- d) What is the probability that it will crash three times in a period of 4 months?

$$P(x) = \frac{(4)^x e^{-4}}{x!}$$

$$\frac{(1)^x e^{-1}}{1!}$$

Example 9:



A customer care center receives on average 2.5 calls every hour.

- a) What is the probability that it will receive at most 4 calls every hour?
- b) What is the probability that it will receive at least 5 calls every hour?



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Mean and Variance



Mean:

$$\mu = \lambda$$

For a Poisson random variable, the variance and mean are the same!

Variance and Standard Deviation

$$\sigma^2 = \lambda$$

$$\sigma = \sqrt{\lambda}$$

where λ = expected number of occurrences in a given experiment.



Poisson approximation To Binomial distribution :

When n is large and p is small, Binomial probabilities can be approximated by Poisson's distribution with mean $\lambda=np$.

The approximation is more accurate if $n \geq 100$ and as long as $np \leq 10$

Example 10:



It is known that 5% of the books bound at a certain bindery have defective bindings. Find the probability that 2 of 100 books bound by this bindery will have defective bindings using

- (a) The formula for the binomial distribution;
- (b) The Poisson approximation to the binomial distribution

Solution:



- (a) Here $x = 2$, $n = 100$, and $p = 0.05$. using the formula for the binomial distribution, we get

$$b(2; 100, 0.05) = \binom{100}{2} (0.05)^2 (0.95)^{98} = 0.081$$

- (b) Substituting $x = 2$ and $\lambda = np = 100(0.05) = 5$ into the formula for the Poisson distribution, we get

$$f(2; 5) = \frac{5^2 \cdot e^{-5}}{2!} = 0.084$$

Example 11:



Given that a switch board of a consultant's office receives on the average 0.6 calls per minute, find the probabilities that

- (a) in a given minute there will be at least 1 call;
- (b) in a 4-minute interval there will be at least 3 calls.

Solution:



a) Here,

$$\lambda = 0.6 \text{ and } t = 1 \therefore p(x, \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

$$p(x, \lambda t) = \frac{e^{-0.6} (0.6)^x}{x!}$$

b)

$$P(X \geq 1) = 1 - P(X < 1) = 1 - P(0)$$

$$= 1 - e^{-0.6} = 0.4512$$

$$\lambda = 0.6 \text{ and } t = 4 \therefore p(x, \lambda t) = \frac{e^{-2.4} (2.4)^x}{x!}$$

$$P(X \geq 3) = 1 - P(X < 3) = 1 - [P(0) + P(1) + P(2)]$$

$$= 0.4303$$

Poisson Distribution: Applications



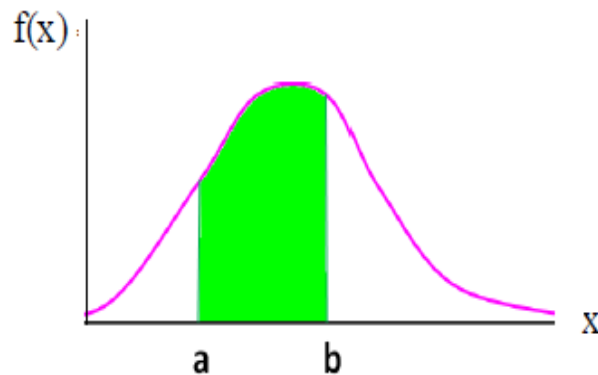
- Arrivals at queuing systems:
 - ✓ Airports : people, airplanes, automobiles, baggage
 - ✓ Banks: people, loan applications
 - ✓ computer file servers : read and write operations
- Defects in manufactured goods:
 - ✓ number of defects per 1,000 feet of extruded copper wire
 - ✓ number of blemishes per square foot of painted surface
 - ✓ number of errors per typed page

RECALL: Continuous Random Variables



A continuous random variable can assume any value in an interval on the real line or in a collection of intervals.

The probability of the random variable assuming a value within some given interval from x_1 to x_2 is defined to be the area under the graph of the **probability density function** between x_1 and x_2



Example:

- Height of students in a class
- Amount of ice tea in a glass
- Change in temperature throughout a day
- Price of a car in next year

Continuous Probability Distributions:



Normal Distribution:

Normal distribution is a general distribution. Any information such as heights, weights, lengths, widths, sizes, profits, prices, wages, time etc. containing sufficiently large data with mean and variance follows normal distribution.

The distribution of errors of repeated measurements follows normal distribution.

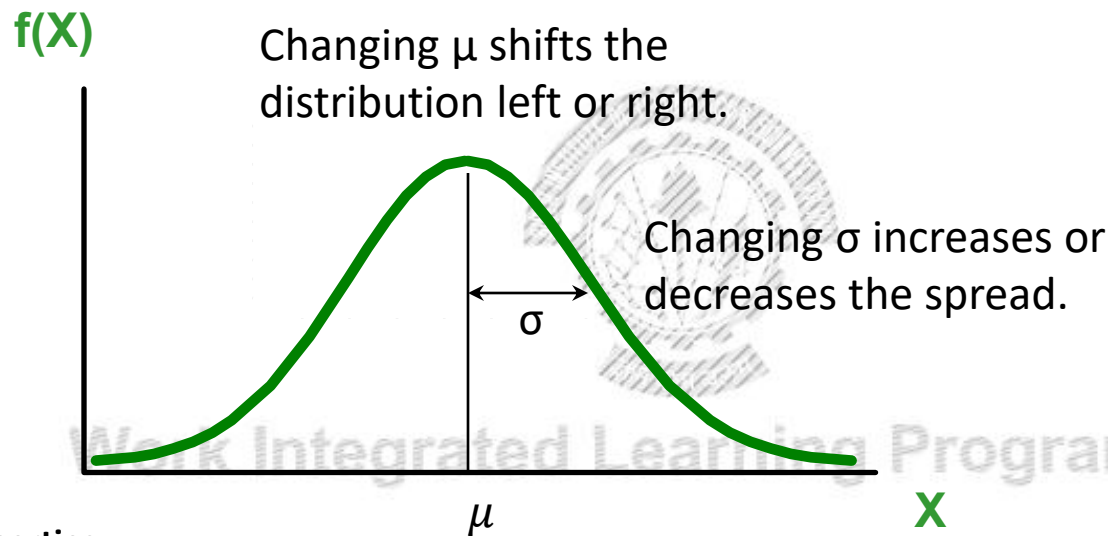
Normal Distribution



A continuous random variable X which assumes all possible values in the entire real space, i.e., $-\infty < x < \infty$ is said to follow normal distribution with two parameters such as mean μ and variance σ^2 if it has the probability density function given by

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \begin{matrix} -\infty < x < \infty \\ -\infty < \mu < \infty, \sigma > 0 \end{matrix}$$

Normal Distribution

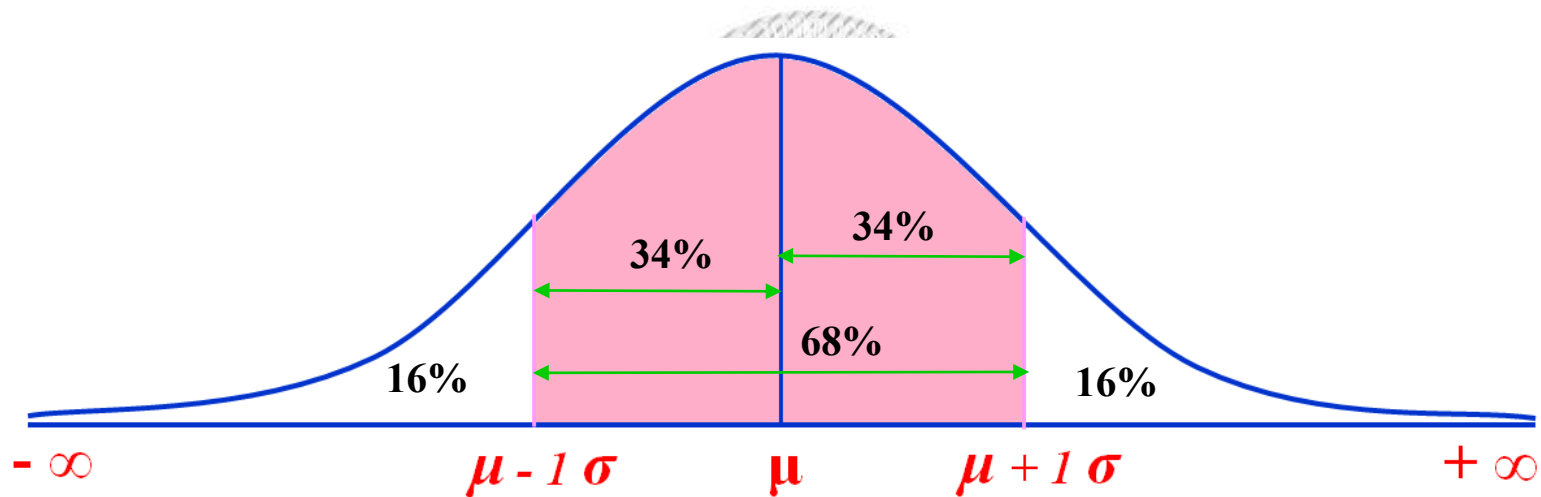


Properties:

1. Normal curve is bell shaped and symmetric about the mean
2. Mean = Mode = Median
3. Total area under normal curve is equal to 1
4. Normal curve approaches but never touches the x axis as it extends farther and farther away from the mean

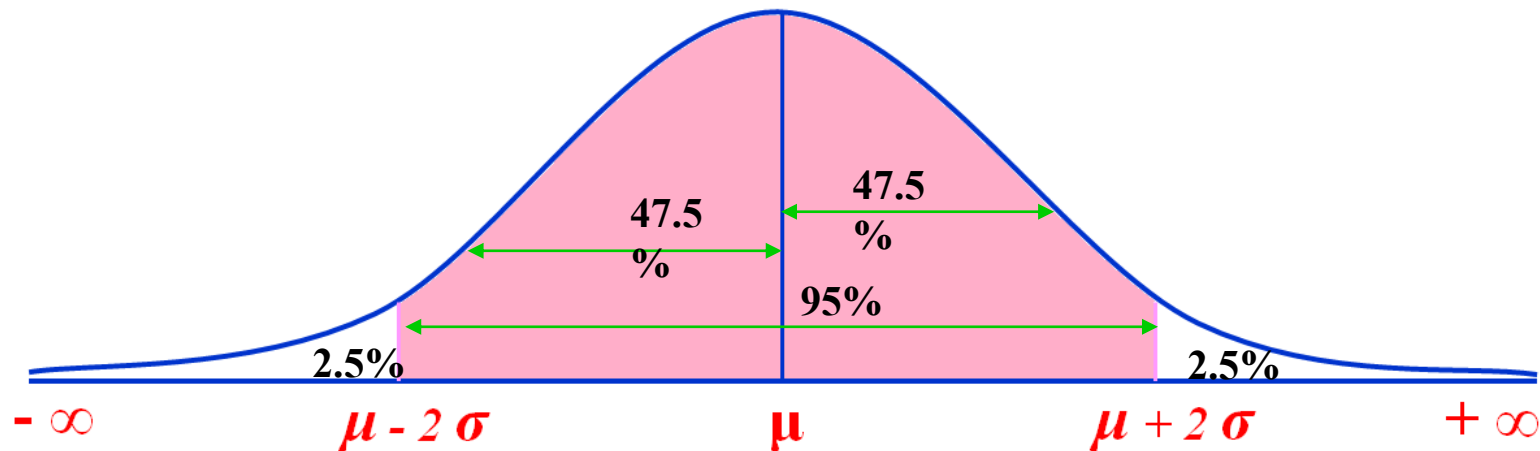
Area under normal curve

Area under the curve: Mean $\pm$ 1SD



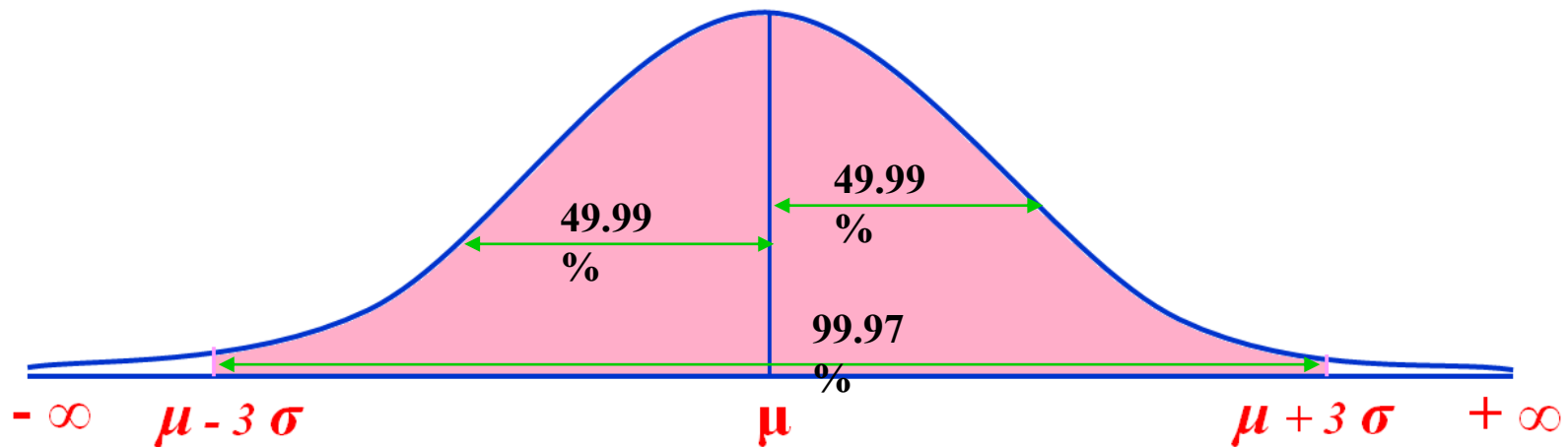
Area under normal curve

Area under the curve: $\text{Mean} \pm 2\text{SD}$



Area under normal curve

Area under the curve: Mean $\pm$ 3SD



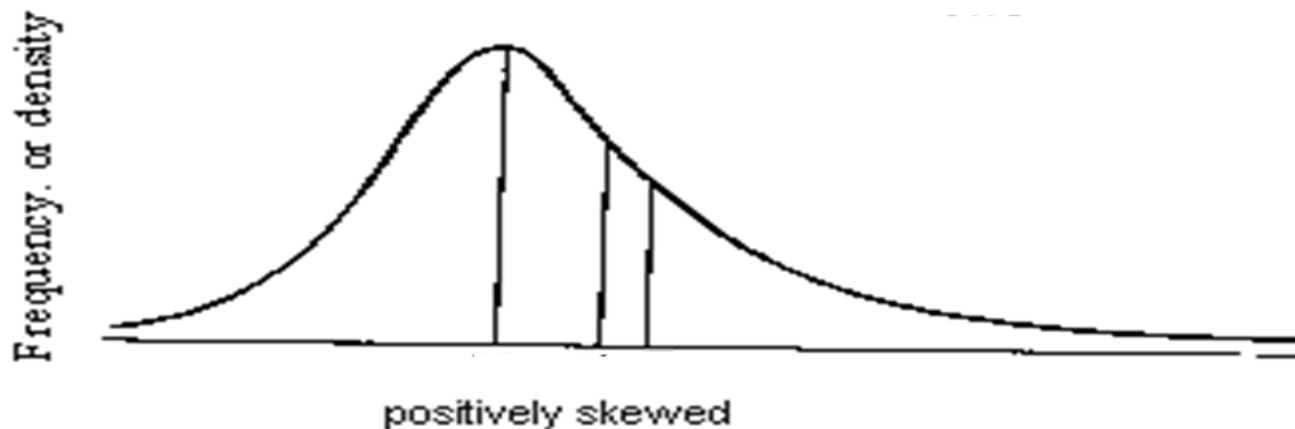
Skewness

Negatively skewed: The left tail is longer; the mass of the distribution is concentrated on the right of the figure. The distribution is said to be left-skewed

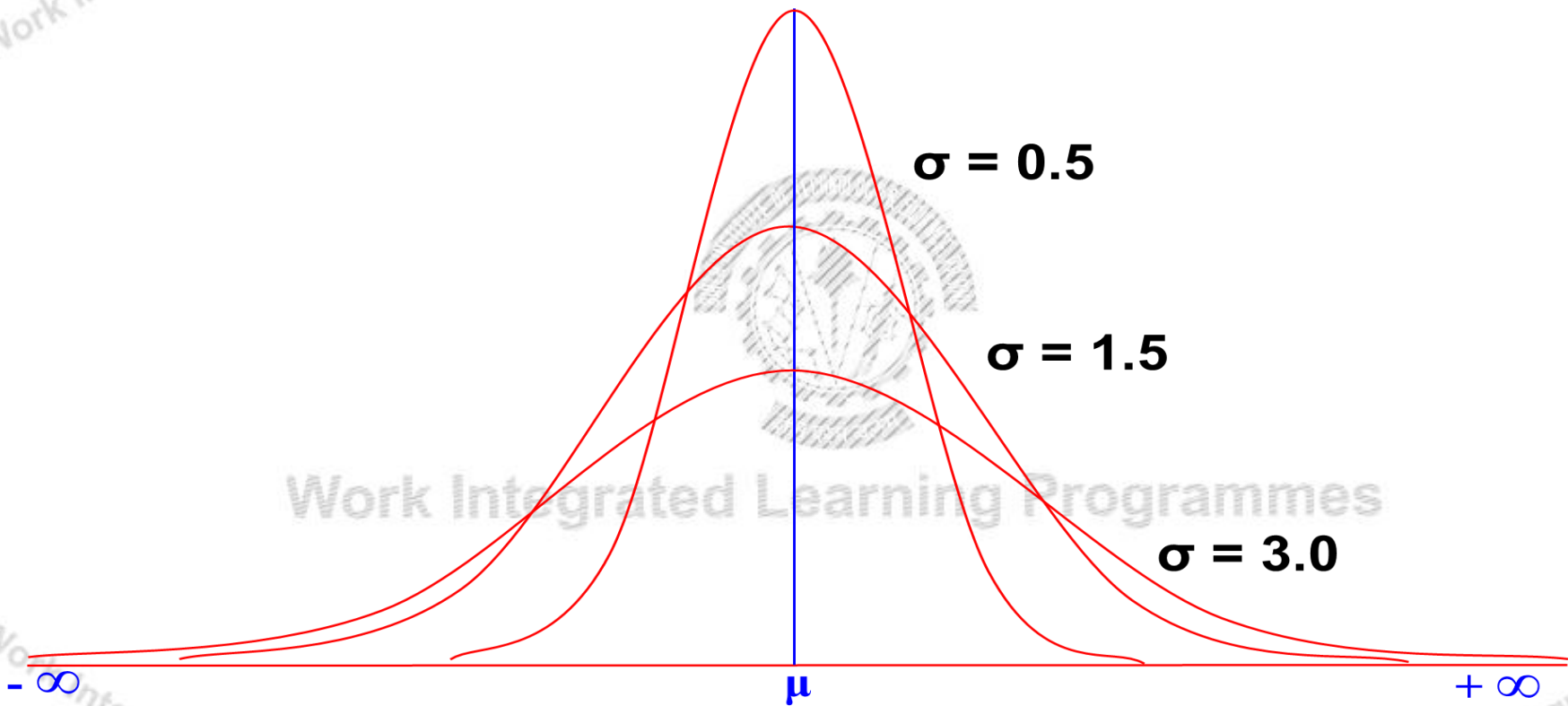


Skewness

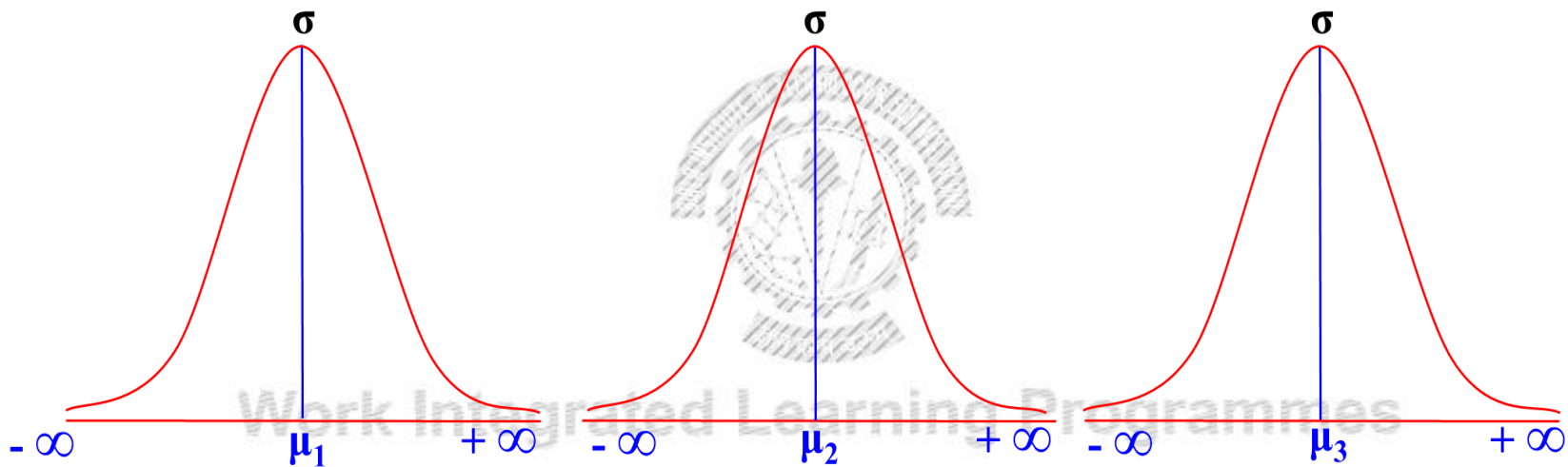
Positively skewed: The right tail is longer; the mass of the distribution is concentrated on the left of the figure. The distribution is said to be right-skewed



The Normal Distribution



The Normal Distribution



Normal Distribution



$$E(X)=\mu = \int_{-\infty}^{+\infty} x \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\text{Var}(X)=\sigma^2 = \left(\int_{-\infty}^{+\infty} x^2 \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \right) - \mu^2$$

Standard Deviation(X)= σ

No matter what μ and σ are, the area between $\mu-\sigma$ and $\mu+\sigma$ is about 68%; the area between $\mu-2\sigma$ and $\mu+2\sigma$ is about 95%; and the area between $\mu-3\sigma$ and $\mu+3\sigma$ is about 99.7%. Almost all values fall within 3 standard deviations.

Why Standard Normal Distribution?

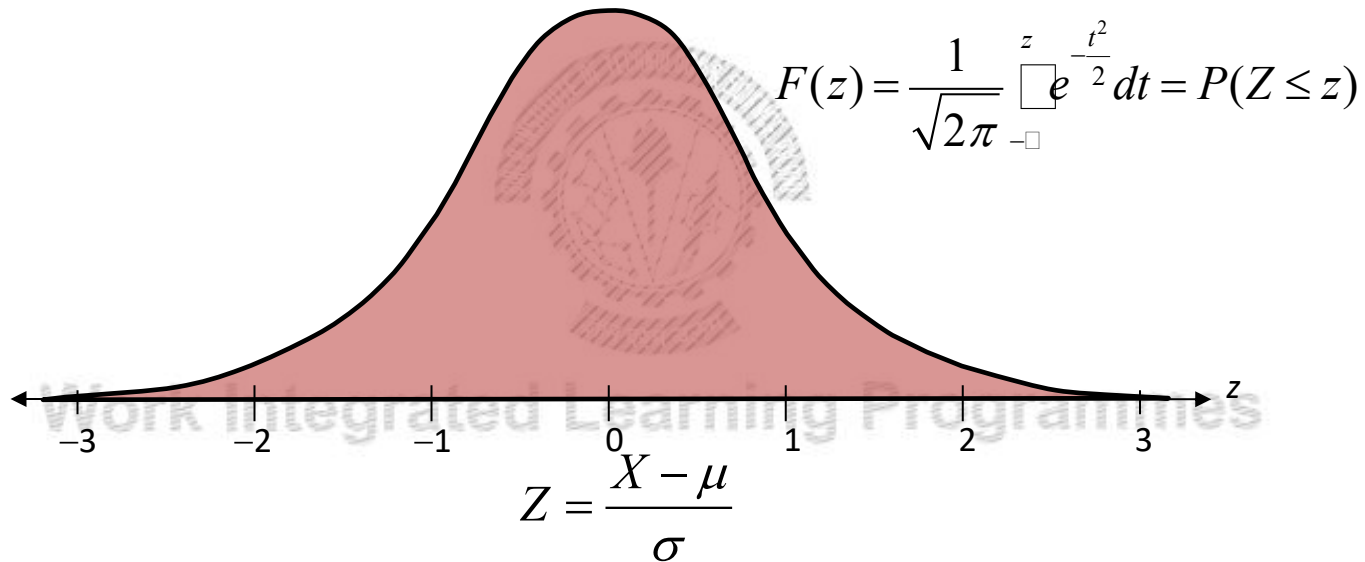
	x	$X - \mu_x$	$(X - \mu_x)^2$	$Z = \frac{X - \mu_x}{\sigma_x}$	$(Z - \mu_z)^2$
	1	-2	4	-1.42	2.01
	2	-1	1	-0.71	0.50
	3	0	0	0.00	0.00
	4	1	1	0.71	0.50
	5	2	4	1.42	2.01
Sum	15	0	10	0	5.03
Mean	$\mu_x = 3$			$\mu_z = 0$	
SD	$\sigma_x = 1.41$			$\sigma_z = 1.00$	

$X \sim N(\mu, \sigma^2) \rightarrow Z \sim N(0, 1)$. This is true for any data size

Standard Normal Distribution



The **standard normal distribution** is a normal distribution with a mean of 0 and a standard deviation of 1.



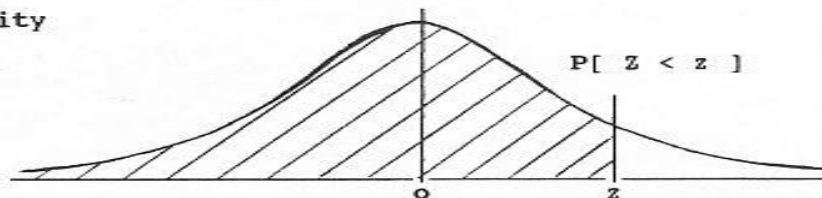
All normal distributions can be converted into the standard normal curve by subtracting the mean and dividing by the standard deviation

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9874	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
z	3.00	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90
P	0.9986	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

Problem 12:



If a random variable has the standard normal distribution, find the probability that it will take on a value

a) less than 1.75 = $P(Z < 1.75) = F(1.75) = 0.9599$

b) less than -1.25 = $P(Z < -1.25) = F(-1.25) = 1 - F(1.25) = 0.1056$

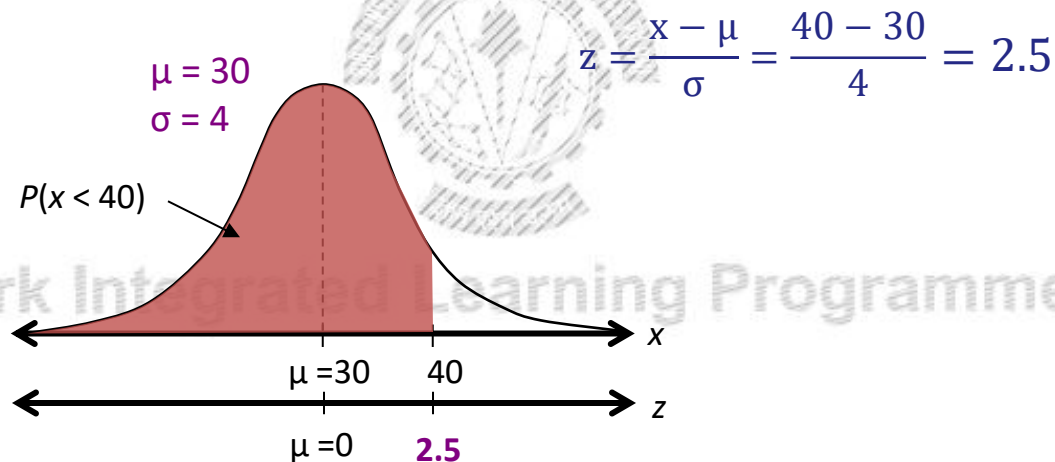
c) greater than 2.06 = $P(Z > 2.06) = 1 - F(2.06) = 0.0197$

d) greater than -1.82 = $P(Z > -1.82) = 1 - F(-1.82) = 1 - 0.0344 = 0.9656$

$P(Z > -1.82) = 1 - F(-1.82) = 1 - (1 - F(1.82)) = F(1.82) = 0.9656$

Problem 13:

Scores on a particular test are normally distributed with a mean of 30 and standard deviation of 4. What is the probability of anyone scoring less than 40?

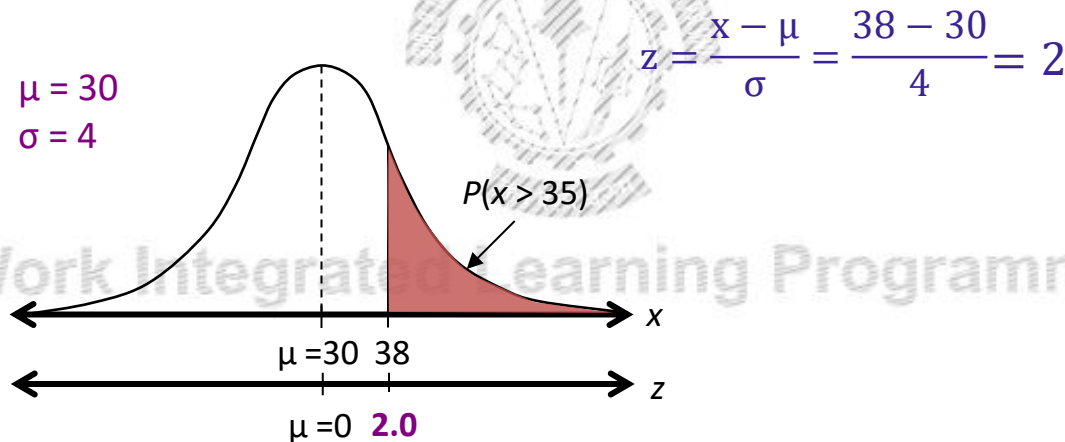


$$P(x < 40) = P(z < 2.5) = 0.9938$$

The probability that a student receives a test score less than 90 is 0.9938.

Problem 14:

Scores on a particular test are normally distributed with a mean of 30 and standard deviation of 4. What is the probability of anyone scoring greater than 38?

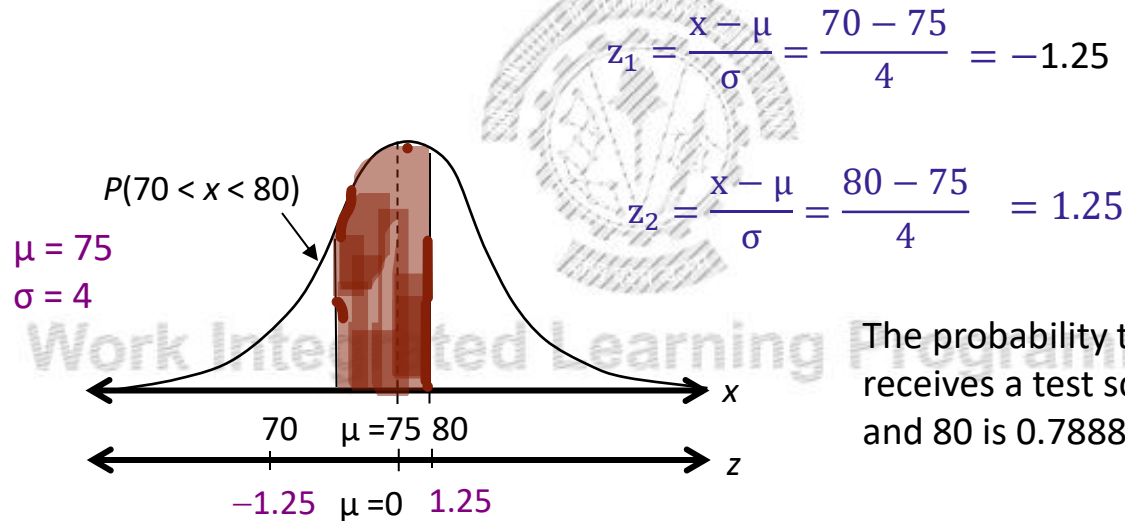


$$P(x > 38) = P(z > 2) = 1 - P(z < 2) = 1 - 0.9773 = 0.0227$$

The probability that a student receives a test score greater than 38 is 0.0227

Problem 15:

The average on a statistics test was 75 with a standard deviation of 4. If the test scores are normally distributed, find the probability that a student receives a test score between 70 and 80.



$$\begin{aligned}
 P(70 < x < 80) &= P(-1.25 < z < 1.25) = F(1.25) - F(-1.25) \\
 &= 0.8944 - 0.1056 = 0.7888
 \end{aligned}$$

Problem 16:

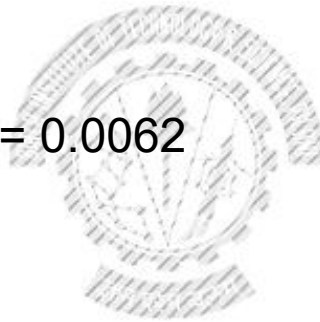


The length of life of an instrument produced by a machine has a normal distribution with a mean of 12 months and a standard deviation of 2 months.

Find the probability that an instrument produced by this machine will last

- a) less than 7 months.
- b) between 7 and 12 months.

Solution: a) $P(x < 7) = P(z < -2.5) = 0.0062$



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b) $P(7 < x < 12) = P(-2.5 < z < 0) = 0.4938$

Problem 17:



Butterfly-style valves used in heating and ventilating industries have a high flow coefficient. Flow coefficient can be modeled by a normal distribution with mean $496 C_v$ and standard deviation $25C_v$. Find the probability that a valve will have a flow coefficient of

a) at least $450C_v$

$$P(X > 450) = P[(450 - 496) / 25] = P(Z > -1.84) = 1 - F(-1.84) = 1 - 0.0329 = .9671$$

b. between 445.5 and $522C_v$

$$\begin{aligned} P(445.5 < X < 522) &= P(-2.02 < Z < 1.04) = F(1.04) - F(-2.02) = 0.8508 - 0.0217 \\ &= 0.8291 \end{aligned}$$

Normal Distribution



Change in Mean determines the shift in the distribution

Change in the deviation determines the spread of the data points

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

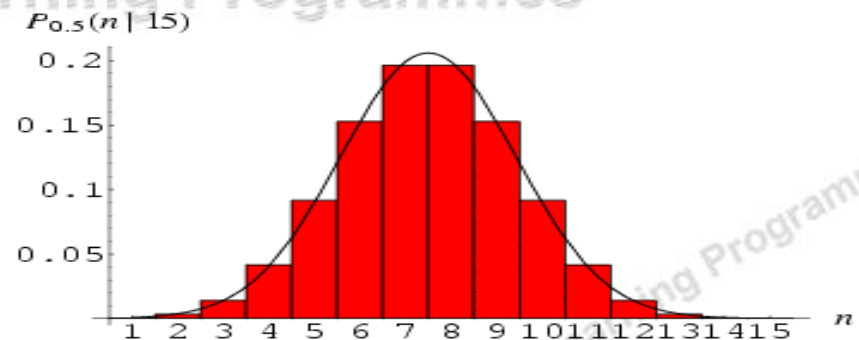
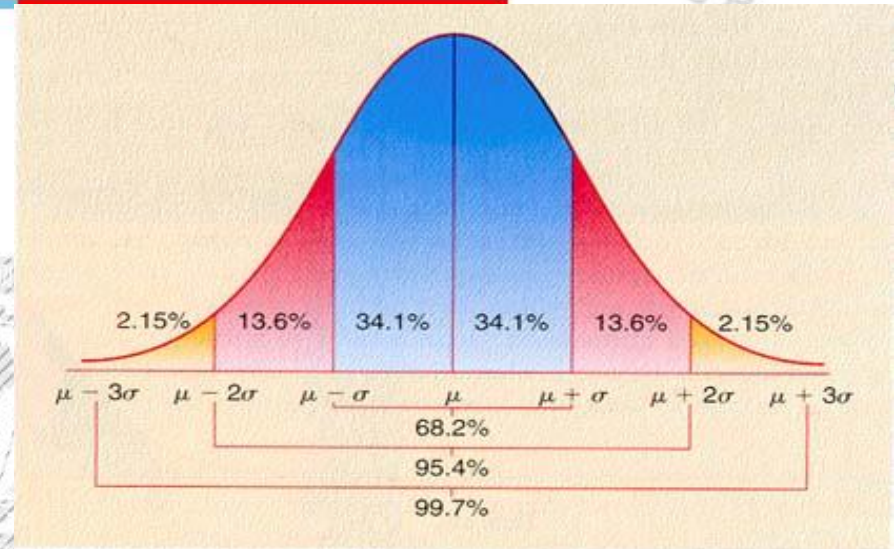
μ = mean

σ = standard deviation

π = 3.14159

e = 2.71828

This is a bell shaped curve with different centers and spreads depending on μ and σ



Normal Approximation



The normal distribution is used to approximate the binomial distribution when it would be impractical to use the binomial distribution to find a probability.

Normal Approximation to a Binomial Distribution

If $np \geq 15$ and $nq \geq 15$, then the binomial random variable x is approximately normally distributed with mean

$$\mu = np$$

and standard deviation

$$\sigma = \sqrt{npq}.$$

Problem 18:



Decided whether the normal distribution to approximate x may be used in the following examples.

1. In a survey of 8 to 18-year-old heavy media users in the United States, 47% said they get fair or poor grades (C's or below). You randomly select forty-five 8 to 18-year-old heavy media users in the United States and ask them whether they get fair or poor grades.

$$np = 45(0.47) = 21.15$$

$$nq = 45(0.53) = 23.85.$$

2. Thirty-six percent of people in the United States own a dog. You randomly select 25 people in the United States and ask them if they own a dog.



$$np = (25)(0.36) = 9$$

$$nq = (25)(0.64) = 16$$

Because np is not greater than 15, the normal distribution may NOT be used.

Problem 19:



Thirty-one percent of the seniors in a certain high school plan to attend college. If 50 students are randomly selected, find the probability that less than 14 students plan to attend college by normal approximation.

$$np = (50)(0.31) = 15.5$$

$$nq = (50)(0.69) = 34.5$$

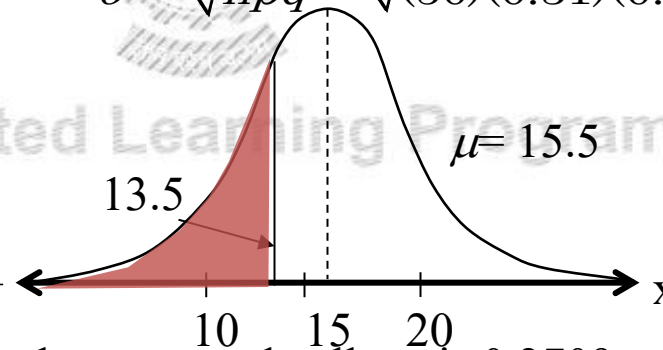
$$P(x < 13.5) = P(z < -0.61) = 0.2709$$

Correction for continuity

$$z = \frac{x - \mu}{\sigma} = \frac{13.5 - 15.5}{3.27} = -0.61$$

The variable x is approximately normally distributed with $\mu = np = 15.5$ and

$$\sigma = \sqrt{npq} = \sqrt{(50)(0.31)(0.69)} = 3.27.$$



The probability that less than 14 plan to attend college is 0.2709.

Problem 20:



If 62% of all clouds seeded with silver iodide show spectacular growth, What is the probability that among 40 clouds seeded with silver iodide at most 20 will show spectacular growth by normal approximation?

Soln: Here $n = 40$, $p = 0.62$, $q = 0.38$,
 $\mu = np = 24.8$, $\sigma^2 = npq = 9.424$, $\sigma = 3.0699$
 $np \geq 15$ and $nq \geq 15$

Therefore we will use normal approximation to binomial distribution.

$$P(x \leq 20.5) = ?$$

$$\text{When } x = 20.5, \quad Z = \frac{x - \mu}{\sigma} = \frac{20.5 - 24.8}{3.0699} = -1.40$$

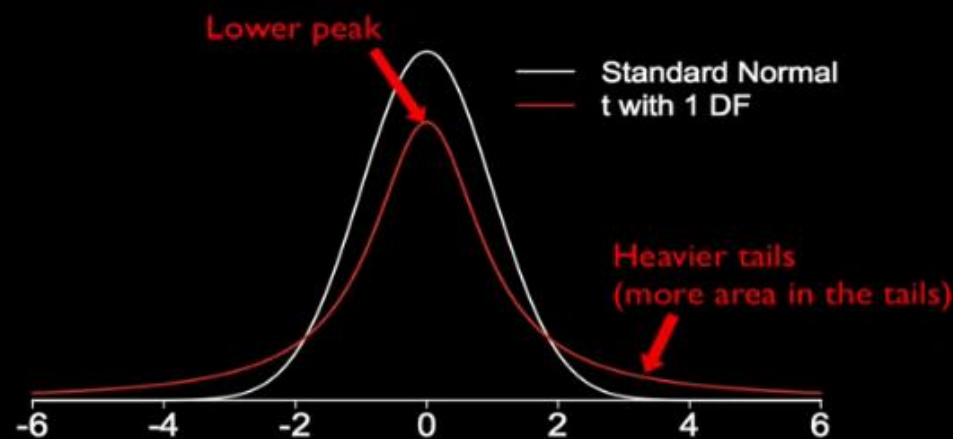
$$\begin{aligned} \therefore P(x \leq 20.5) &= P(Z \leq -1.40) \\ &= F(-1.40) = 0.808 \end{aligned}$$



Self - study

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Introduction to t-distribution



— Standard Normal
— t with 1 DF

Heavier tails
(more area in the tails)

-6 -4 -2 0 2 4 6

As the degrees of freedom increase, the t distribution tends toward the standard normal distribution

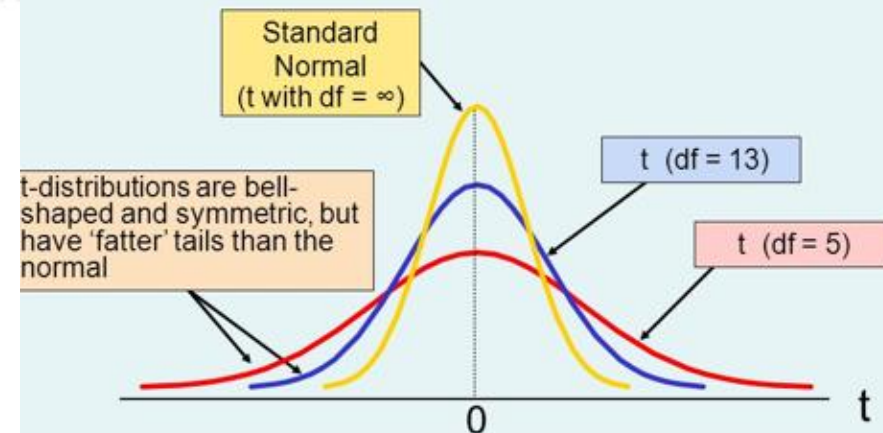
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The pdf of the t distribution with ν degrees of freedom:

$$f(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

for $-\infty < t < \infty$

Note: $t \rightarrow Z$ as n increases



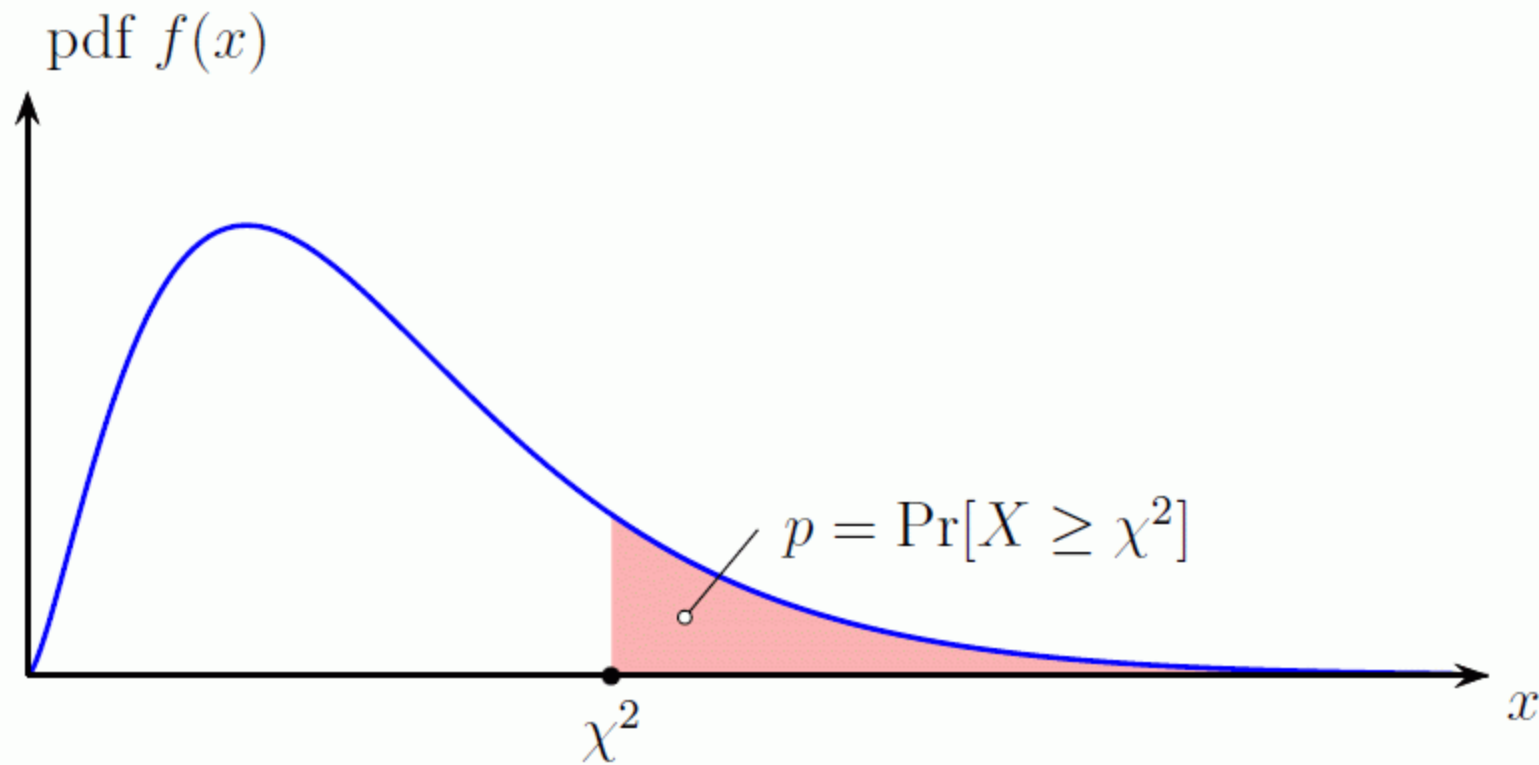
Introduction to Chi-Square(χ^2) distribution

- Chi-Square distribution pdf is given by

$$f(x) = \begin{cases} \frac{1}{2^{\vartheta/2} \Gamma(\frac{\vartheta}{2})} x^{\frac{\vartheta}{2}-1} e^{-\frac{x}{2}} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

where ϑ is the parameter of distribution, also known as degrees of freedom.

- Introduction to Chi-Square(χ^2) distribution





Introduction to Chi- Square(χ^2) distribution

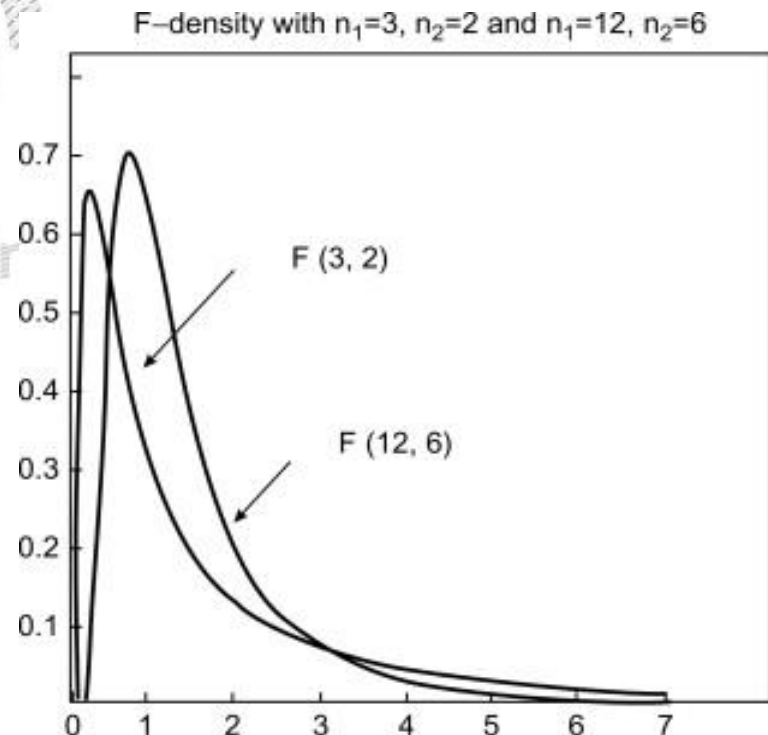
- ❖ Chi-Square distribution curve is not symmetrical, and hence not a normal curve.
- ❖ Chi-Square varies from 0 to ∞ (Curve lies entirely in first quadrant).
- ❖ It depends only on degrees of freedom ν .
- ❖ It is very important in estimation and hypothesis testing.
- ❖ It is used in sampling distributions of the sample variance, analysis of variance.
- ❖ It is used as a measure of goodness of fit.

Introduction to F-distribution (Sampling distribution of the ratio of two sample variances)

- The pdf for a random variable $X \sim F(n_1, n_2)$ is given by

$$f(x) = \frac{\Gamma\left(\frac{n_1 + n_2}{2}\right)}{\Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right)} \left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} x^{\frac{n_1}{2}-1} \left(1 + \frac{n_1}{n_2} x\right)^{-\frac{n_1+n_2}{2}}, x > 0$$

- F- distribution curve lies entirely in first quadrant.
- The F- curve depends not only on the two parameters ν_1 and ν_2 but also on the order in which they are stated.



Introduction to F-distribution

(Sampling distribution of the ratio of two sample variances)



- ❖ The F-distribution was **developed by Fisher to study the behavior of two variances from random samples taken from two independent normal populations.**
- ❖ In applied problems we may be interested in knowing whether the population variances are equal or not, based on the response of the random samples.
- ❖ If S_1^2 and S_2^2 are variances of independent random sample of size n_1 and n_2 from a normal populations with variances σ_1^2 and σ_2^2 , then

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2}$$

which follows F –distribution with $\nu_1 = n_1 - 1$ and $\nu_2 = n_2 - 1$ d.o.f.

Introduction to t-distribution (Sampling distribution of mean when σ unknown)



The t-distribution, also known as the **Student's t-distribution**, is a type of probability distribution that is similar to the normal distribution with its bell shape but has heavier tails.

It is used for estimating population parameters for **small sample sizes when**

t - Distribution When σ is unknown and $n < 30$.

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} \text{ with } (n-1) \text{ d.o.f.}$$

Where sample S.D. is calculated by formula

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Practice Problems



1. The number of customers arriving at a cafeteria at an average rate of 0.3 per minute.
 - (a) Find the probability that exactly 2 customers arrive in a 10-minute span.
 - (b) Find the probability that 2 or more customers arrive in a 10-minute span.
 - (c) Find the probability that exactly one customer arrives in a 5-minute span and one customer arrives in the next 5-minute span.

Practice Problems



2. In a factory, 8% of all machines break down at least once a year. Use the Poisson approximation to the binomial distribution to determine the probabilities that among 25 machines (randomly chosen in the factory):
- (a) 5 will break down at least once a year;
 - (b) at least 4 will break down once a year:
 - (c) anywhere from 3 to 8, inclusive, will break down at least once a year.

Practice Problems



3. The average number of trucks arriving on any one day at a truck depot in a certain city is known to be 12. what is the probability that on a given day fewer than nine trucks will arrive at this depot?

Practice Problems



4. A certain kind of sheet metal has on the average, five defects per 10 square feet. If we assume a Poisson distribution, what is the probability that a 15-square-foot sheet of the metal will have at least six defects?

Practice Problems



5. The number of flaws in a fiber optic cable follows a poisson process with an average of 0.6 per 100 feet.
- a) Find the probability of exactly 2 flaws in 200 foot cable.
 - b) Find the probability of exactly 1 flaw in first 100 feet and exactly 1 flaw in the second 100 feet.

Practice Problems



6. Suppose that on an average 1 out of 10000 houses catch fire in a year in a district. If there are 2000 houses in the district, find the probability that exactly 5 house will catch fire during that year.

Practice Problems



7. Find the probability that at most 5 defective fuses will be found in a box of 200 fuses if experiences shows that 2% of such fuses are defective. [Ans 0.785]
8. Suppose one percent of the items made by a machine are defective. In a sample of 100 items, find probability that sample contains: (a) no defective item (b) 1 defective item (c) 3 or more defective items.[Ans (a) 0.368 (b) 0.368 (c) 0.080]
9. Suppose 2 percent of the people on the average are left-handed. Find the probability 3 or more left handed among 100 people.[Ans 0.325]
10. Wireless sets are manufactured with 25 soldered joints, each on the average 1 joint in 500 defective. How many sets can be expected to the free from defective joints in a consignment of 10000 sets? [Ans:0.9512]



Thank you

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